

Labb 2 Ethical Hacking

By Samuel Melin

2.1

a) The Dmitry tool is a tool used for information-gathering. It gathers public and network-level data.

Some examples of options you have with the tool is:

- -W

Is a whois lookup where you can collect domain owner, name servers, contact information, registration dates etc.

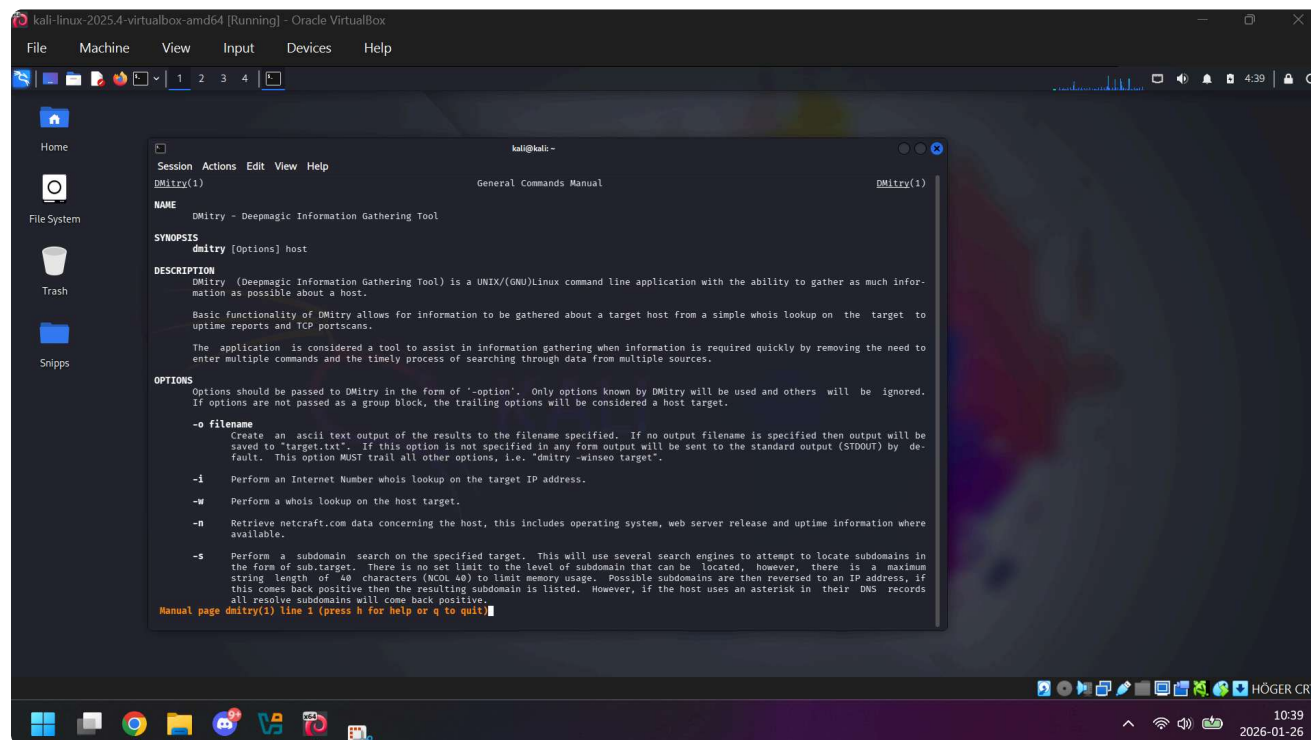
- -S

Can be used for finding subdomains such as www,mail,ftp, dev etc.

- -e

Is used for scraping emails from web pages and search engine results.

The tool offers a lot of other options that can be used for collection information.

A screenshot of a Kali Linux virtual machine window. The terminal is open, displaying the help menu for the Dmitry tool. The menu includes sections for NAME, SYNOPSIS, DESCRIPTION, and OPTIONS. The OPTIONS section lists various flags like -o, -i, -w, -n, and -s with their respective functions. The terminal window has a dark theme and a sidebar on the left with icons for Home, File System, Trash, and Snippets. The bottom status bar shows the time as 10:39 and the date as 2026-01-26.

```
kali@kali:~$ dmitry --help
NAME
  Dmitry - Deepmagic Information Gathering Tool

SYNOPSIS
  dmitry [Options] host

DESCRIPTION
  Dmitry (Deepmagic Information Gathering Tool) is a UNIX/GNU/Linux command line application with the ability to gather as much information as possible about a host.

  Basic functionality of Dmitry allows for information to be gathered about a target host from a simple whois lookup on the target to uptime reports and TCP portscans.

  The application is considered a tool to assist in information gathering when information is required quickly by removing the need to enter multiple commands and the timely process of searching through data from multiple sources.

OPTIONS
  Options should be passed to Dmitry in the form of '-option'. Only options known by Dmitry will be used and others will be ignored. If options are not passed as a group block, the trailing options will be considered a host target.

  -o filename
    Create an ascii text output of the results to the filename specified. If no output filename is specified then output will be saved to 'target.txt'. If this option is not specified in any form output will be sent to the standard output (STDOUT) by default. This option MUST trail all other options, i.e. 'dmitry -winesse target'.

  -i
    Perform an Internet Number whois lookup on the target IP address.

  -w
    Perform a whois lookup on the host target.

  -n
    Retrieve netcraft.com data concerning the host, this includes operating system, web server release and uptime information where available.

  -s
    Perform a subdomain search on the specified target. This will use several search engines to attempt to locate subdomains in the form of sub.target. There is no set limit to the level of subdomain that can be located, however, there is a maximum string length of 40 characters (NCOL 40) to limit memory usage. Possible subdomains are then reversed to an IP address, if this comes back positive then the resulting subdomain is listed. However, if the host uses an asterisk in their DNS records all resolve subdomains will come back positive.

Manual page dmitry(1) line 1 (press h for help or q to quit)
```

b)

I ran the command "dig @ns1.du.se du.se (http://du.se) AXFR"

This interrogated the server which returned a complete zone information. This also indicated that the server allows for zone transfers from the current network but a lot of the information is encrypted:

```
h23samum@maggie:~$ dig @ns1.du.se du.se AXFR

; <<> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<> @ns1.du.se du.se AXFR
(2 servers found)
;; global options: +cmd
du.se.                300    IN      SOA     ns1.du.se. domainmaster.du.se. 2026012101 14400 3600 2592000 3600
du.se.                300    IN      RRSIG   SOA 8 2 300 20260220072501 20260121072501 21838 du.se. eXDzh8fSMrvZBynVC6KwuWuDuoMUTnEasO8IChP
JE/msAC8g2Zno6UzX 1+30r7rzDck4VWYr/Key/Fwfqsb60lp7Y5zu15ttLf5HXXNnXnlheE CtRTvVLfJ24PaFh38K09YyI5cLywDfp4yT4jjDE9LKOTAB/XwfIMVwEt +CzABWmFoz
FuA5kVPeNBARF8vttx2+PfM0vEE5ArhPLLzj0ZxHezE/6A rSrnC5FF6uD5XiFPwpgFgBTWxj55IwqIrzi1Ujp5P0Xg48y5XcLktwjot qq/WcFdW5/cMrpy3TPeZbAk13iH/DDd2d087Qd
K82ocwhSVYrrRUjqz3 0AWRjPl19p30+tKRXuDjkQ14pUAsDmi0pZDbbSZFiaLaInx8+shYcKCW W3mUqCzDSXS2TqT3uePqu3rfRWKWi+XXEt+l3g71rfyroqf35ht40gkp 2Czpo6/JX
+Xl4tZbAXkf01RsV0+gbr0IggG0D055gvESW47/kHVPeaaSh 5FFsdt5nuIUcuu50CzZD7V0/74e6X1JgqE6bLBMuRnNZ0hjC6foHZJo0 htX1atGsQiw/R08j7tshBAdD5NmBJA8sAJ4Li
LGJRw4XV105uxycbAgR LSbYfrXH8X1ppKYbhIT7I1bSqTg/aSsBxHD2eAOXeEum0Du7Gg0sUwVt oHXMS0fRj58=
du.se.                300    IN      NS      ns1.du.se.
du.se.                300    IN      NS      ns2.du.se.
du.se.                300    IN      NS      sunic.sunet.se.
du.se.                300    IN      RRSIG   NS 8 2 300 20260220072501 20260121072501 21838 du.se. tewqXqnkjBuHH/PBFP9T1qtHQmrjvLKvBNsDCk9E
Z1DnTLFrIQ6KoX7v 8SXQoM6Kbx9jRa4vcKHAtoJAtnBnLfSlwe8DoMeUsnmU5zxQmEHtEt LUOpdow34gmSrjg1YQvMDWOk0tw+4d3TDGuDS0SpwwglYJIarW6vDiTN wnJJ2Aa+zs+
hN7xQewNq9l3/LaLYpqlmLzHyJImyYJaCl8vKHLWk9CMmJ NbUrVdp2Siuls+N6oQrvbtPCh93hPR0sNy+Q9C1tTuP/NE6Wd4vYKiiI Xu0YehLPwN5thQSeKXHA5FDfx7sM95ZduURwJX
8MXLvg/dMP1Q0KdmS MZg10b5yQKHH/rkJC1fF4LhJZIGmWbWgslVvCe79hRp4rFWrcWGzRpDC TAwXm51pXme5wDw0bcMFT/cRgXjCzjiboYKVSrC2prKebile1DSi2WY OsQBEDq9Wt
WcstygnSgvz0dqCK0Hpo0kfWH/LzwdBxBnXjzAcwpoLSc OBL7XZAgz+P293ZLRlJYzIgp88JYWKAA8B7d9LIIF8Lbv/xH0tLFoBc01a JetC4T+zBHV041otQe4MQGqXir8tb6rIoJsZZ+
QLqTf594jaDvs3tLfX WLCH65q6X12RsbvRGA+3WLqkLH86IP264c9AuL5+iQAD6uYsKyc4nLQX E6a8GiWVC+E=
du.se.                300    IN      A       130.243.57.42
du.se.                300    IN      RRSIG   A 8 2 300 20260220072501 20260121072501 21838 du.se. oNBROhm9xUAHDobl7duz0W90/icu00XAh0Gvtxbkj
PBk40ERV4UV8iEn haEDjWwbeHI6ukJwIJ4WKKi5IJzeo7WdQbCayJLDH9WDpgoGwLsQArfd IiiTeCHPjxWtrB7Xp2lFuv/VKK5ecy12mxgMipyTf0xJ/WLF4y4F+eP7 WE8UvrKpVm70
f1bAcPtQoaDZp1v6Doxa+0txGDuIL4i0wZMrw6y5f0C Yee8Bka90XHx143uvUTZiLHE//7lBr6kbW65xJeUggY7ZxALQpZ0szWU JUoYctspotoBn0vKtgUw0v0Qa2j538umebbhHVVLMb
RvsIE/9FnzmArYAM 43Au5WuaqtciJt2LQHR6gMXqTUXg/utBy+fndk5LmSiET5YcRoPy6JR 2a0RyUdQhRggGAZAIwLCX4DPyqPpFTdEUxHgmui4qLCQJktkbWMxz01j QhK6jCZ1RhK
NafymAJW7jxPHZoLHLUC3aLSsaWsgvomJM9uTt36qByVF RG89kR5sduUZHWEHWHetIw6ZyIO18rJl9hBuuo0/ug/3uizVCNjjufj S8y31zSBPh08wzEPwqX3IQymUCsMaIbtqc9Fh2
XQXpIwtd3Kxaea33Z 6RRD0Fid4vtB+IZJY12aWcSSWqcy6X2AM5OYpMGjube67E1ZhVpDaIT7 MQ6jj7nPiME=
du.se.                300    IN      MX      10 mailfilter-ng-1.sunet.se.
du.se.                300    IN      MX      10 mailfilter-ng-3.sunet.se.
du.se.                300    IN      MX      10 mailfilter-ng-4.sunet.se.
du.se.                300    IN      MX      20 mailfilter-ng-2.sunet.se.
du.se.                300    IN      RRSIG   MX 8 2 300 20260220072501 20260121072501 21838 du.se. DuIxpiscOlvdM5cyDTcm3LDtOxLsM8MiZBdH04kl
01vzm01tMLNIntXc NtBMXVGyOqVlsjkZL2AFSGlduPzP9W7sbej6E9067267GLzObk1E0hdl /BI/G4Z1P9FDRjHD+OnDQrqTZjmB/FnVCMYRn07H30i6huh2V26X/faf C+3MTubzTOh
pDd/1ZQqhloanXfsFMDcq8EcihMdEw+QbTYrUVcLCJGWx 4+3Znf+OwyKwbujbyZcw/F9hTyleK4Jp4sPq99rn/sB5HCCJkAoNTw44 JbJD7KTe4/x3IOU0mH0S6H4QEWFcTd//RYXHWPO
iAXfeGohZ2VJfnFMT AxdFHbicyILBrzaWV6DW1A8Ynnfv2Zdug9p/L4/us25mP7LJX8QVRcyGu k4+ztzCI1J4EyKdc8kf0GowBRd3fTzhJMd/mdjjlocZCzhDc6C3rY5Q 0EV61XWqgS
cxgH77QcheW8VxU57jx1dYugjuYXHjFfuVLF908r6bqiH n7Cb035GcHv9W36AEY8sYijqBeB5/r646BW/3y13burDY2kZz2rCHwRN GiscLEFSitLaHdEJSRIFURGqbNPNP5K9ttIT099
```

c) First i used "dig du.se (http://du.se) NS".

This gives me the information about the name servers:

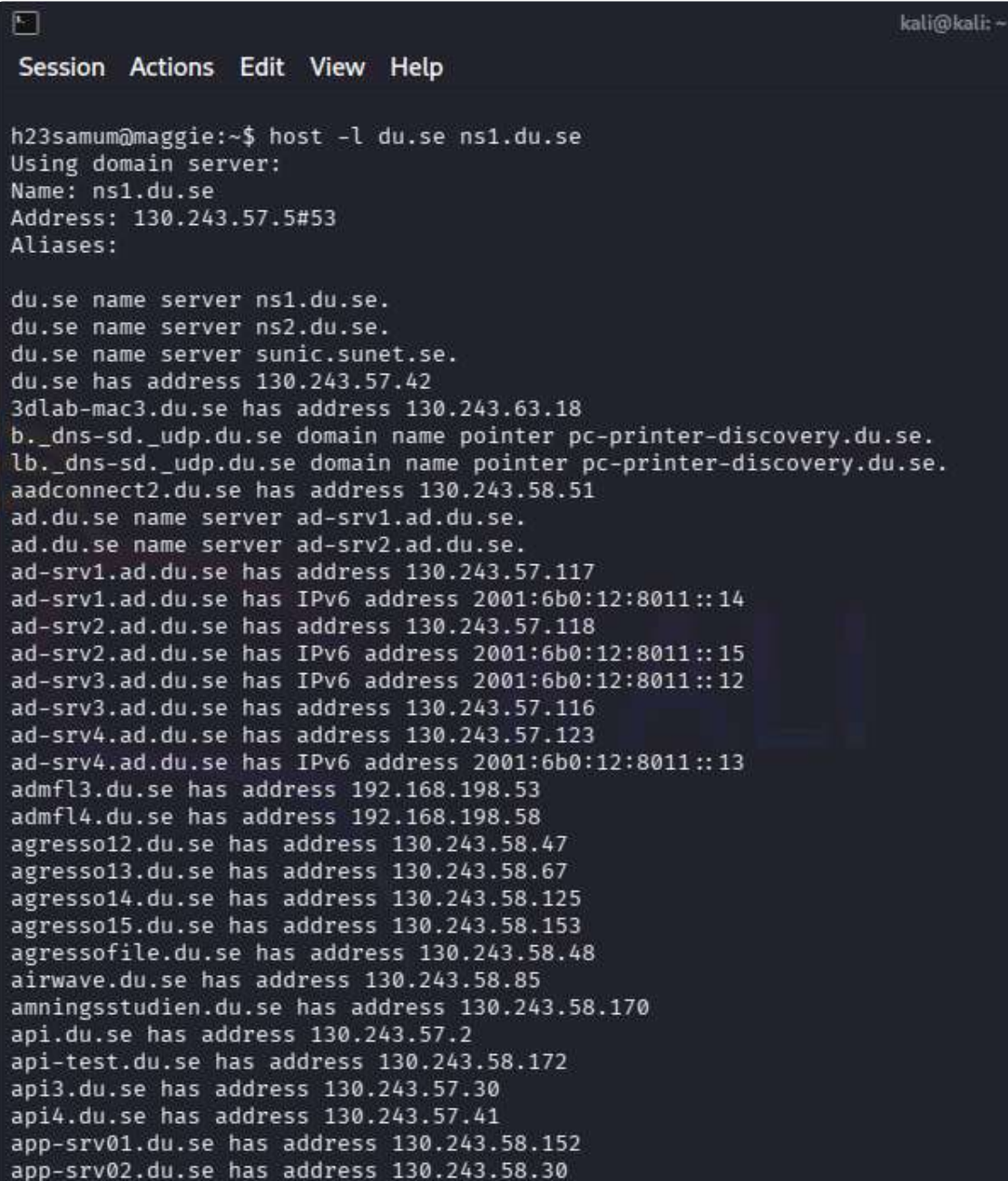
```
kali@kali: ~  
Session Actions Edit View Help  
h23samum@maggie:~$ dig du.se NS  
  
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> du.se NS  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 30847  
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 7  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 65494  
;; QUESTION SECTION:  
;du.se.                                IN      NS  
  
;; ANSWER SECTION:  
du.se.                300      IN      NS      ns2.du.se.  
du.se.                300      IN      NS      ns1.du.se.  
du.se.                300      IN      NS      sunic.sunet.se.  
  
;; ADDITIONAL SECTION:  
ns1.du.se.           300      IN      A        130.243.57.5  
ns2.du.se.           300      IN      A        130.243.32.5  
sunic.sunet.se.      53250    IN      A        192.36.125.2  
ns1.du.se.           300      IN      AAAA     2001:6b0:12::10  
ns2.du.se.           300      IN      AAAA     2001:6b0:12:801::10  
sunic.sunet.se.      53250    IN      AAAA     2001:6b0:7::2  
  
;; Query time: 1 msec  
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)  
;; WHEN: Mon Jan 26 16:02:33 CET 2026  
;; MSG SIZE rcvd: 228  
  
h23samum@maggie:~$
```

After this i tried "nslookup" on ns1.du.se (http://ns1.du.se).

This is simple and clear view if you want quick information, rather using the "dig" command when you get the list with nameservers.

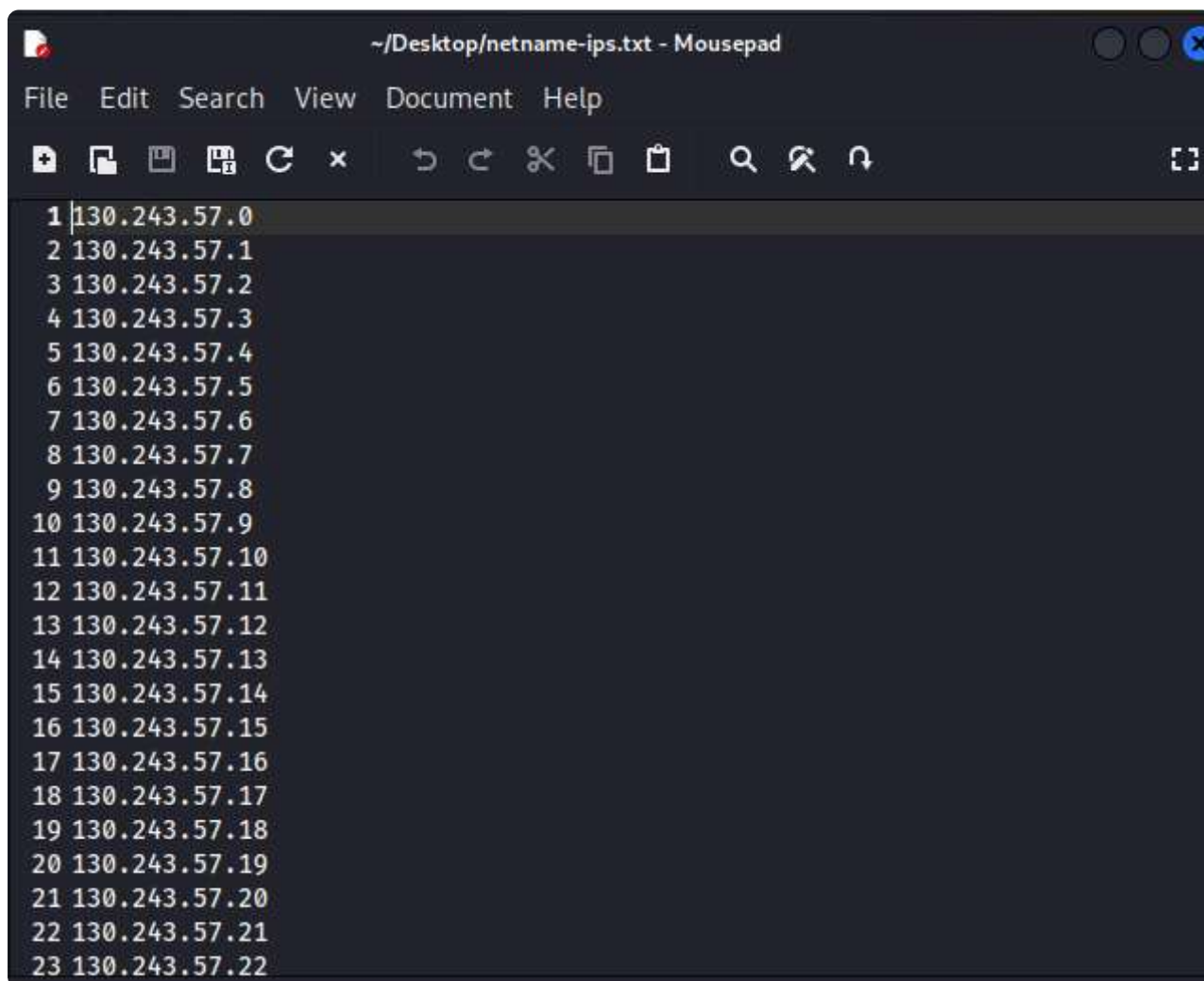
```
Session Actions Edit View Help  
h23samum@maggie:~$ nslookup  
> ns1.du.se  
Server:          127.0.0.53  
Address:         127.0.0.53#53  
  
Non-authoritative answer:  
Name:   ns1.du.se  
Address: 130.243.57.5  
Name:   ns1.du.se  
Address: 2001:6b0:12::10  
>
```

Using the "host" command and combining it with the authoritative name server I got a complete zone, including internal infrastructure:



```
kali@kali: ~  
Session Actions Edit View Help  
  
h23samum@maggie:~$ host -l du.se ns1.du.se  
Using domain server:  
Name: ns1.du.se  
Address: 130.243.57.53  
Aliases:  
  
du.se name server ns1.du.se.  
du.se name server ns2.du.se.  
du.se name server sunic.sunet.se.  
du.se has address 130.243.57.42  
3dlab-mac3.du.se has address 130.243.63.18  
b._dns-sd._udp.du.se domain name pointer pc-printer-discovery.du.se.  
lb._dns-sd._udp.du.se domain name pointer pc-printer-discovery.du.se.  
aadconnect2.du.se has address 130.243.58.51  
ad.du.se name server ad-srv1.ad.du.se.  
ad.du.se name server ad-srv2.ad.du.se.  
ad-srv1.ad.du.se has address 130.243.57.117  
ad-srv1.ad.du.se has IPv6 address 2001:6b0:12:8011::14  
ad-srv2.ad.du.se has address 130.243.57.118  
ad-srv2.ad.du.se has IPv6 address 2001:6b0:12:8011::15  
ad-srv3.ad.du.se has IPv6 address 2001:6b0:12:8011::12  
ad-srv3.ad.du.se has address 130.243.57.116  
ad-srv4.ad.du.se has address 130.243.57.123  
ad-srv4.ad.du.se has IPv6 address 2001:6b0:12:8011::13  
admfl3.du.se has address 192.168.198.53  
admfl4.du.se has address 192.168.198.58  
agresso12.du.se has address 130.243.58.47  
agresso13.du.se has address 130.243.58.67  
agresso14.du.se has address 130.243.58.125  
agresso15.du.se has address 130.243.58.153  
agrossofile.du.se has address 130.243.58.48  
airwave.du.se has address 130.243.58.85  
amningsstudien.du.se has address 130.243.58.170  
api.du.se has address 130.243.57.2  
api-test.du.se has address 130.243.58.172  
api3.du.se has address 130.243.57.30  
api4.du.se has address 130.243.57.41  
app-srv01.du.se has address 130.243.58.152  
app-srv02.du.se has address 130.243.58.30
```

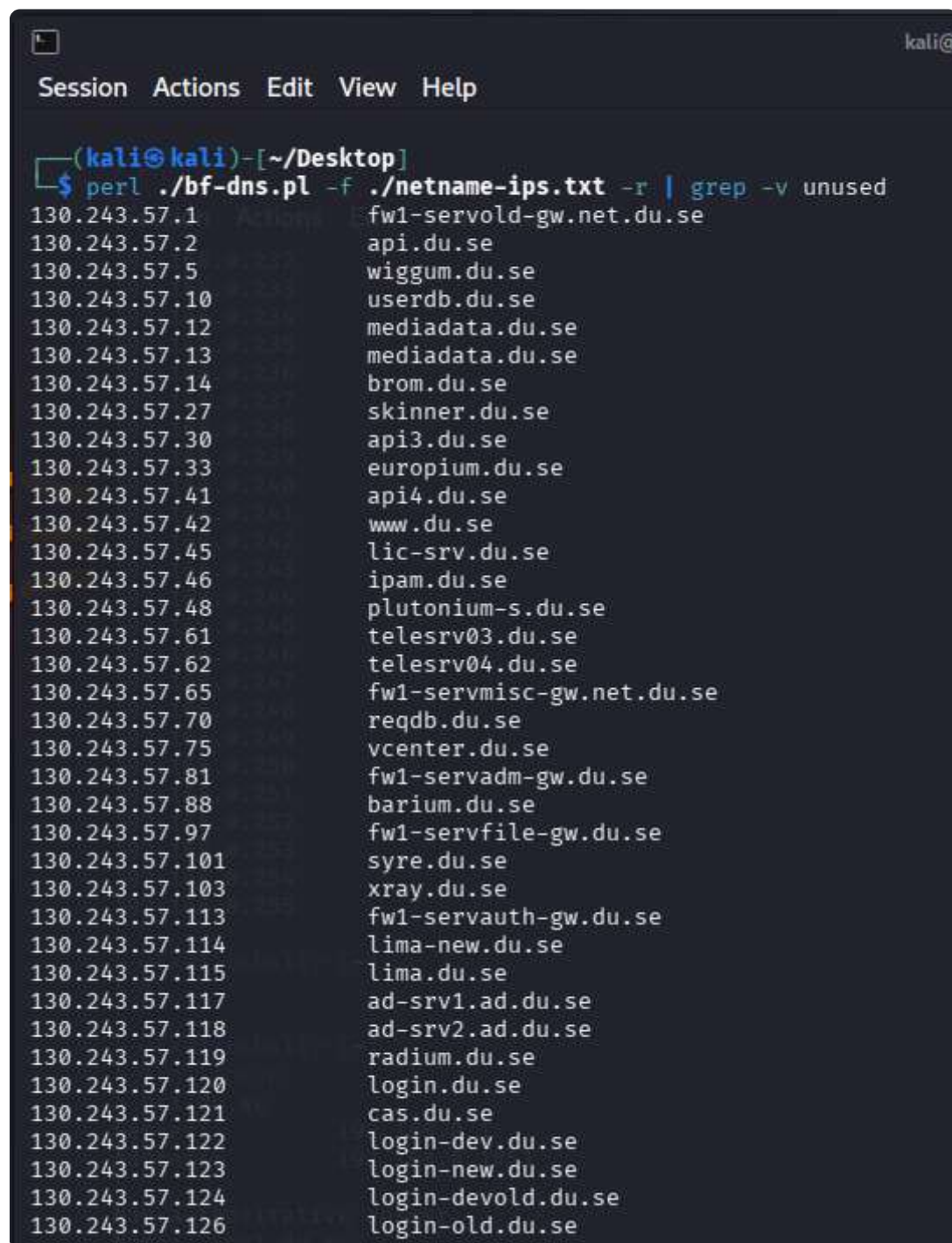

d) Yes it works together with the bf-dns.pl (<http://bf-dns.pl>) script. First i used "netenum" and generated a txt file:



The screenshot shows a text editor window titled "~/Desktop/netname-ips.txt - Mousepad". The window has a menu bar with "File", "Edit", "Search", "View", "Document", and "Help". Below the menu bar is a toolbar with various icons for file operations and editing. The main text area contains a list of 23 IP addresses, each preceded by a number from 1 to 23. The IP addresses are in the format 130.243.57.x, where x ranges from 0 to 22.

```
1 130.243.57.0
2 130.243.57.1
3 130.243.57.2
4 130.243.57.3
5 130.243.57.4
6 130.243.57.5
7 130.243.57.6
8 130.243.57.7
9 130.243.57.8
10 130.243.57.9
11 130.243.57.10
12 130.243.57.11
13 130.243.57.12
14 130.243.57.13
15 130.243.57.14
16 130.243.57.15
17 130.243.57.16
18 130.243.57.17
19 130.243.57.18
20 130.243.57.19
21 130.243.57.20
22 130.243.57.21
23 130.243.57.22
```

After that i used the script together with perl command which resolved the hostnames:



The screenshot shows a Kali Linux terminal window with a dark background. At the top, there is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. Below the menu bar, the prompt is '(kali@kali)-[~/Desktop]'. The command executed is '\$ perl ./bf-dns.pl -f ./netname-ips.txt -r | grep -v unused'. The output is a list of IP addresses and their corresponding hostnames. The IP addresses are listed on the left, and the hostnames are listed on the right. The hostnames are mostly .du.se domains.

```
(kali@kali)-[~/Desktop]
$ perl ./bf-dns.pl -f ./netname-ips.txt -r | grep -v unused
130.243.57.1      fw1-servold-gw.net.du.se
130.243.57.2      api.du.se
130.243.57.5      wiggum.du.se
130.243.57.10     userdb.du.se
130.243.57.12     mediadata.du.se
130.243.57.13     mediadata.du.se
130.243.57.14     brom.du.se
130.243.57.27     skinner.du.se
130.243.57.30     api3.du.se
130.243.57.33     europium.du.se
130.243.57.41     api4.du.se
130.243.57.42     www.du.se
130.243.57.45     lic-srv.du.se
130.243.57.46     ipam.du.se
130.243.57.48     plutonium-s.du.se
130.243.57.61     telesrv03.du.se
130.243.57.62     telesrv04.du.se
130.243.57.65     fw1-servmisc-gw.net.du.se
130.243.57.70     reqdb.du.se
130.243.57.75     vcenter.du.se
130.243.57.81     fw1-servadm-gw.du.se
130.243.57.88     barium.du.se
130.243.57.97     fw1-servfile-gw.du.se
130.243.57.101    syre.du.se
130.243.57.103    xray.du.se
130.243.57.113    fw1-servauth-gw.du.se
130.243.57.114    lima-new.du.se
130.243.57.115    lima.du.se
130.243.57.117    ad-srv1.ad.du.se
130.243.57.118    ad-srv2.ad.du.se
130.243.57.119    radium.du.se
130.243.57.120    login.du.se
130.243.57.121    cas.du.se
130.243.57.122    login-dev.du.se
130.243.57.123    login-new.du.se
130.243.57.124    login-devold.du.se
130.243.57.126    login-old.du.se
```

e) The script performs one reverse DNS lookup per IP address. The IP address is reversed and sends a PTR query to DNS.

If the PTR exists it will print out the hostnames. If it doesn't exist the IP is listed as "unused" and removed from the output.

f) I ran the command "patator dns_forward name=FILE0 0=duhosts.txt" and it worked the other way around and resolved the IP addresses for the hostnames:

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Domains
    du.se 284
    ad.du.se 6
    clients.du.se 111
    play.du.se 2
    student.du.se 3
Networks
    130.243.32.5
    130.243.33.248
    130.243.34.x
    130.243.35.x
    130.243.36.x
    130.243.37.x
    130.243.38.x
    130.243.44.x
    130.243.45.x
    130.243.46.x
    130.243.47.x
    130.243.48.x
    130.243.51.x
    130.243.52.x
    130.243.54.x
    130.243.55.x
    130.243.56.66
    130.243.57.x
    130.243.58.x
    130.243.61.x
    130.243.62.18
    130.243.63.x
    2001:6b0:12::10
    2001:6b0:12::20
    2001:6b0:12:4c2::10
    2001:6b0:12:801::10
    2001:6b0:12:8010::20
    2001:6b0:12:8011::14
    2001:6b0:12:8011::15
    2001:6b0:12:8020::11
    2001:6b0:12:8020::13
    2001:6b0:12:8020::14
02:38:34 patator INFO - Hits/Done/Skip/Fail/Size: 743/743/0/0/743, Avg: 71 r/s, Time: 0h 0m 10s
```

g) Some other tools that I found out that can be used for this is dnsrecon which looks simple to use for brut, reverse and zone transfer.

There is several tools that can be used like this but here is an example of dnsrecon

when i tried it:

```
kali@kali: ~/Desktop
Session Actions Edit View Help
2026-01-27T02:50:29.676935-0500 INFO Completed enumeration for domain: du.se

(kali@kali)-[~/Desktop]
$ dnsrecon -d du.se -t brt -D du-prefixes.txt

2026-01-27T02:51:53.161792-0500 INFO Using the dictionary file: /usr/share/dnsrecon/dnsrecon/data/namelist.txt (provided by tool)
2026-01-27T02:51:53.162178-0500 INFO Starting enumeration for domain: du.se
2026-01-27T02:51:53.162517-0500 INFO brt: Performing host and subdomain brute force against du.se...
2026-01-27T02:51:53.434622-0500 INFO A ad.du.se 130.243.57.117
2026-01-27T02:51:53.434803-0500 INFO A ad.du.se 130.243.57.118
2026-01-27T02:51:53.434879-0500 INFO AAAA ad.du.se 2001:6b0:12:8011::14
2026-01-27T02:51:53.434944-0500 INFO AAAA ad.du.se 2001:6b0:12:8011::15
2026-01-27T02:51:53.595979-0500 INFO CNAME agent.du.se telesrv03.du.se
2026-01-27T02:51:53.596190-0500 INFO A telesrv03.du.se 130.243.57.61
2026-01-27T02:51:53.708445-0500 INFO CNAME app.du.se nihonium.du.se
2026-01-27T02:51:53.709403-0500 INFO A nihonium.du.se 130.243.58.8
2026-01-27T02:51:53.841856-0500 INFO CNAME autodiscover.du.se autodiscover.outlook.com
2026-01-27T02:51:53.842049-0500 INFO CNAME autodiscover.outlook.com atod-g2.tm-4.office.com
2026-01-27T02:51:53.842312-0500 INFO CNAME atod-g2.tm-4.office.com autod.ms-acdc-autod.office.com
2026-01-27T02:51:53.842402-0500 INFO A autod.ms-acdc-autod.office.com 40.98.16.11
2026-01-27T02:51:53.842537-0500 INFO A autod.ms-acdc-autod.office.com 132.245.230.21
2026-01-27T02:51:53.842667-0500 INFO A autod.ms-acdc-autod.office.com 52.98.151.88
2026-01-27T02:51:53.842744-0500 INFO A autod.ms-acdc-autod.office.com 40.104.62.200
2026-01-27T02:51:53.842868-0500 INFO CNAME autodiscover.du.se autodiscover.outlook.com
2026-01-27T02:51:53.842988-0500 INFO CNAME autodiscover.outlook.com atod-g2.tm-4.office.com
2026-01-27T02:51:53.843061-0500 INFO CNAME atod-g2.tm-4.office.com autod.ms-acdc-autod.office.com
2026-01-27T02:51:53.843186-0500 INFO AAAA autod.ms-acdc-autod.office.com 2603:1026:900:38::15
2026-01-27T02:51:53.843303-0500 INFO AAAA autod.ms-acdc-autod.office.com 2603:1026:900:3::13
2026-01-27T02:51:53.843385-0500 INFO AAAA autod.ms-acdc-autod.office.com 2603:1026:c11:915::8
2026-01-27T02:51:53.843547-0500 INFO AAAA autod.ms-acdc-autod.office.com 2603:1026:c11:8::8
2026-01-27T02:51:54.104663-0500 INFO CNAME bob.du.se lic-srv.du.se
2026-01-27T02:51:54.104847-0500 INFO A lic-srv.du.se 130.243.57.45
2026-01-27T02:51:54.391764-0500 INFO CNAME catalog.du.se dalarnauniversity-open-vanity.eu-central.catalog.canvaslms.com
2026-01-27T02:51:54.392317-0500 INFO CNAME dalarnauniversity-open-vanity.eu-central.catalog.canvaslms.com catalog-prod-4f07gjn50-19272498
27.eu-central-1.elb.amazonaws.com
2026-01-27T02:51:54.394298-0500 INFO A catalog-prod-4f07gjn50-1927249827.eu-central-1.elb.amazonaws.com 3.127.115.152
2026-01-27T02:51:54.394459-0500 INFO A catalog-prod-4f07gjn50-1927249827.eu-central-1.elb.amazonaws.com 52.59.15.193
2026-01-27T02:51:55.824083-0500 INFO CNAME cr.du.se proxy01.du.se
2026-01-27T02:51:55.824323-0500 INFO A proxy01.du.se 130.243.58.124
```

2.2

a) For TCP connect scan i used "nmap -sT scanme." Ports shown in the image:

```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)-[~/Desktop]
$ nmap -sT 130.243.36.2
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 03:10 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.25 seconds

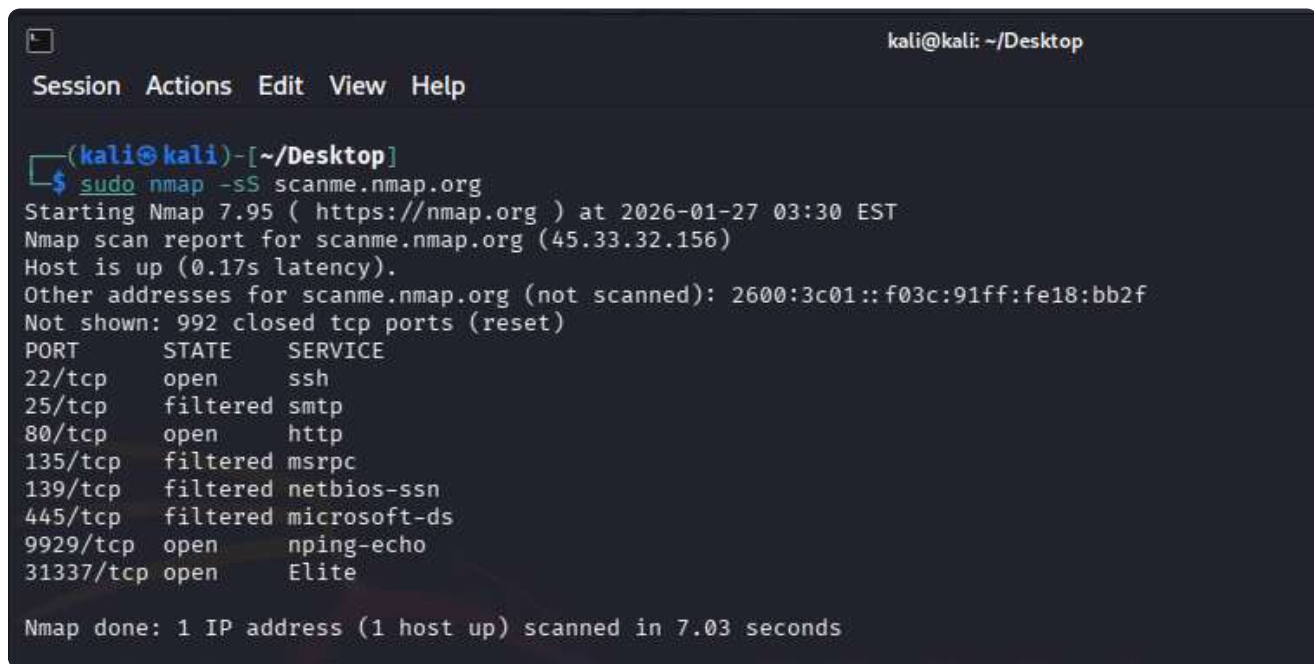
(kali@kali)-[~/Desktop]
$ nmap -sT scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 03:13 EST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.16s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (conn-refused)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered  smtp
80/tcp    open       http
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open       nping-echo
31337/tcp open       Elite

Nmap done: 1 IP address (1 host up) scanned in 4.53 seconds

(kali@kali)-[~/Desktop]
$
```

This shows which ports are open, filtered(can be blocked by firewall). There is no unfiltered and closed ports here.

b) For this task I used the command "nmap -sS scanme.nmap.org (<http://scanme.nmap.org>)". According to the manual, using SYN it never completes the TCP connection which makes it more stealthy to use:



The screenshot shows a terminal window with a dark background. At the top right, it says 'kali@kali: ~/Desktop'. Below that is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal prompt is '(kali@kali)-[~/Desktop]'. The user has entered the command '\$ sudo nmap -sS scanme.nmap.org'. The output shows the Nmap version (7.95), the target IP (45.33.32.156), and a list of open and filtered ports with their corresponding services. The scan took 7.03 seconds.

```
(kali@kali)-[~/Desktop]
$ sudo nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 03:30 EST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.17s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered   smtp
80/tcp    open       http
135/tcp   filtered   msrpc
139/tcp   filtered   netbios-ssn
445/tcp   filtered   microsoft-ds
9929/tcp  open       nping-echo
31337/tcp open       Elite

Nmap done: 1 IP address (1 host up) scanned in 7.03 seconds
```

c) The check UPD ports the command "nmap -sU scanme.nmap.org (<http://scanme.nmap.org>)" was used. As seen in the image the it took some time to perform this scan:

kali@kali: ~/Desktop

Session Actions Edit View Help

```
Stats: 0:01:09 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 7.95% done; ETC: 03:49 (0:13:19 remaining)
Stats: 0:01:12 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 8.35% done; ETC: 03:49 (0:13:10 remaining)
Stats: 0:01:22 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 8.73% done; ETC: 03:51 (0:14:28 remaining)
Stats: 0:01:40 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 10.46% done; ETC: 03:51 (0:14:16 remaining)
Stats: 0:02:26 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 14.38% done; ETC: 03:52 (0:14:35 remaining)
Stats: 0:04:34 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 26.59% done; ETC: 03:52 (0:12:36 remaining)
Stats: 0:11:18 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 64.43% done; ETC: 03:52 (0:06:14 remaining)
Stats: 0:13:02 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 74.29% done; ETC: 03:52 (0:04:31 remaining)
Stats: 0:15:03 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 85.57% done; ETC: 03:52 (0:02:32 remaining)
Stats: 0:15:39 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 89.03% done; ETC: 03:52 (0:01:56 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.16s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed udp ports (port-unreach)
PORT      STATE      SERVICE
68/udp    open|filtered dhcpc
123/udp   open       ntp
135/udp   filtered   msrpc
136/udp   filtered   profile
137/udp   filtered   netbios-ns
138/udp   filtered   netbios-dgm
139/udp   filtered   netbios-ssn
162/udp   open|filtered snmptrap
445/udp   filtered   microsoft-ds
520/udp   open|filtered route

Nmap done: 1 IP address (1 host up) scanned in 1081.13 seconds
```

d) Here I used the command "nmap -A scanme.nmap.org (<http://scanme.nmap.org>)." (you can also use with the option -O):

```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 04:12 EST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.16s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256  96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256  33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-favicon: Nmap Project
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Go ahead and ScanMe!
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Device type: general purpose/router
Running: Linux 5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 5.0 - 5.14, MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 4 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 1723/tcp)
HOP RTT      ADDRESS
1   2.87 ms   192.168.0.1
2   6.41 ms   195.67.144.194
3   6.98 ms   213.64.106.56
4  160.62 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.60 seconds

```

e) From the previous task the conclusion that we can do from this with the results we have based on what nmap can collect is:

- SSH - OpenSSH 6.6.1p1
- HTTP - Apache 2.4.7
- Firewall - Filtered SMB/NetBios
- OS - Linux (Ubuntu) Details 5.0 - 5.14

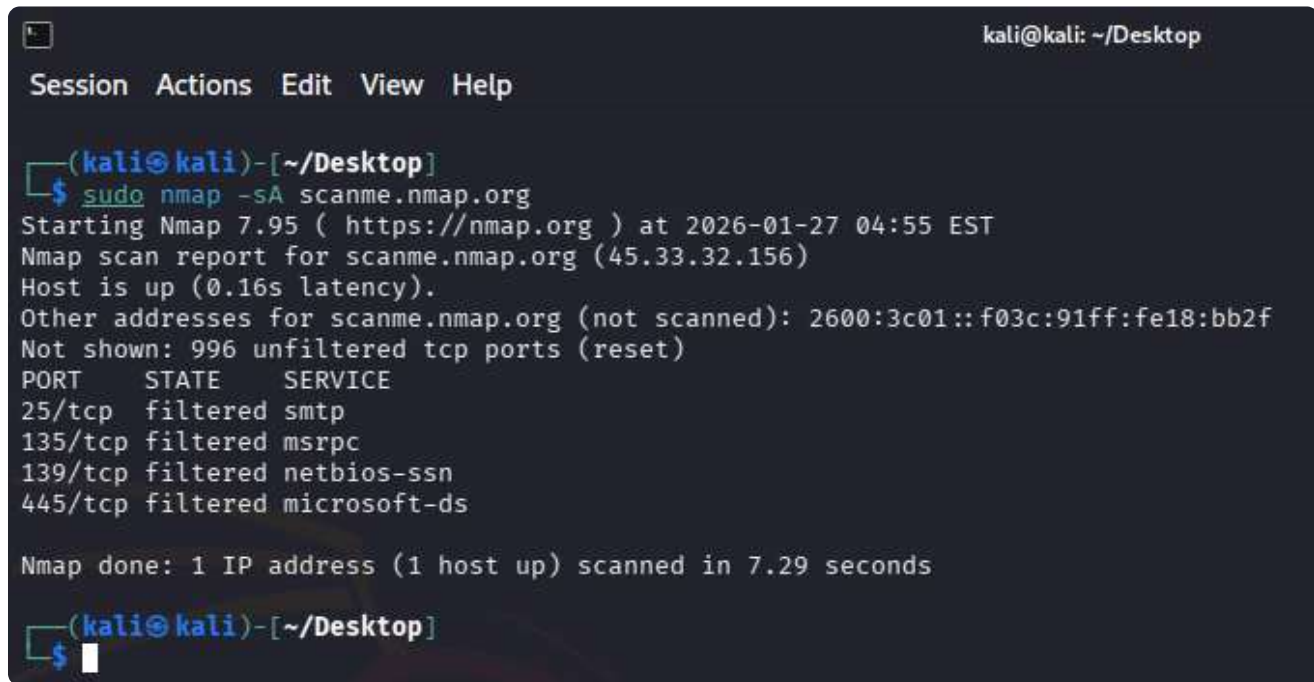
2.3

a) I used the command "nmap -sA scanme.nmap.org (<http://scanme.nmap.org>)." -sA is designed for firewall filtering analysis.

It is fast and in the output you can also see the unfiltered port. You could also use hping3 for manual verification of each port.

b) Here is when I used nmap together with -sA on scanme.nmap.org

(<http://scanme.nmap.org>).



```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)-[~/Desktop]
$ sudo nmap -sA scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-27 04:55 EST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.16s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 unfiltered tcp ports (reset)
PORT      STATE      SERVICE
25/tcp    filtered  smtp
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 7.29 seconds

(kali@kali)-[~/Desktop]
$
```

From this scan we can see that the 996 unfiltered ports respond with RST packets, which means that traffic is allowed through the firewall. These ports can therefore be either open or closed, but they are not filtered.

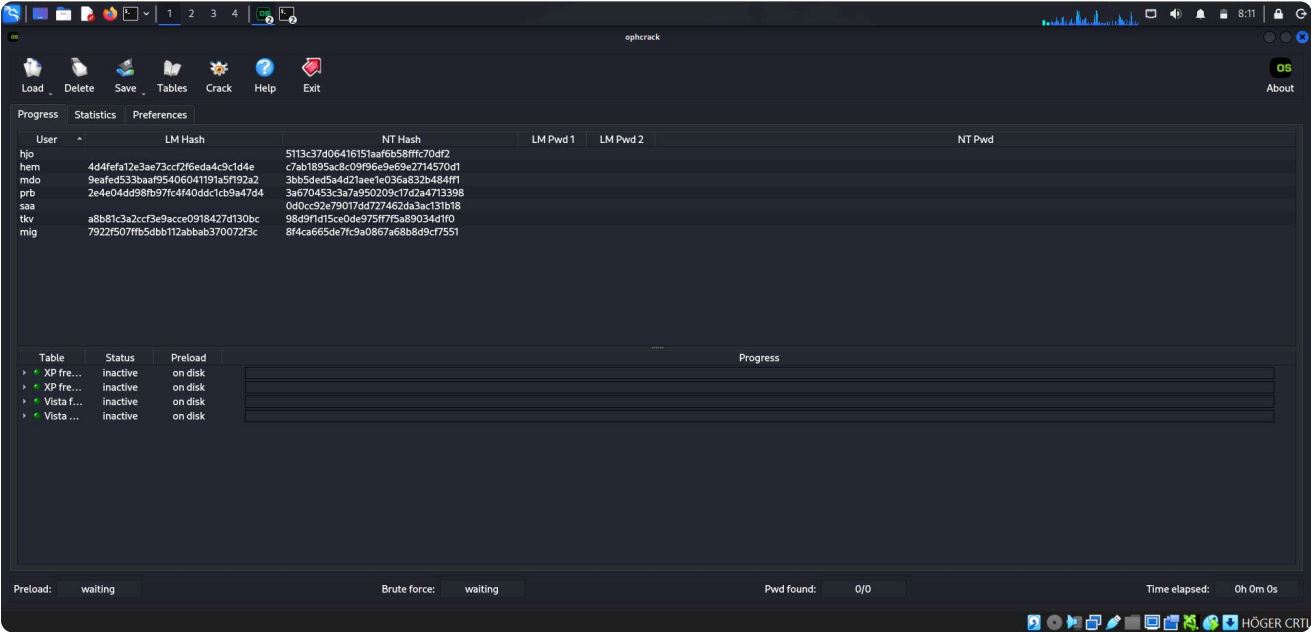
The filtered ports do not respond to ACK packets, indicating that traffic is blocked by a firewall or filtering device.

If the scan were performed from inside the network, fewer ports might be filtered depending on internal firewall rules and user access rights. Therefore, the scan result depends on whether the scanning host is located inside or outside the network.

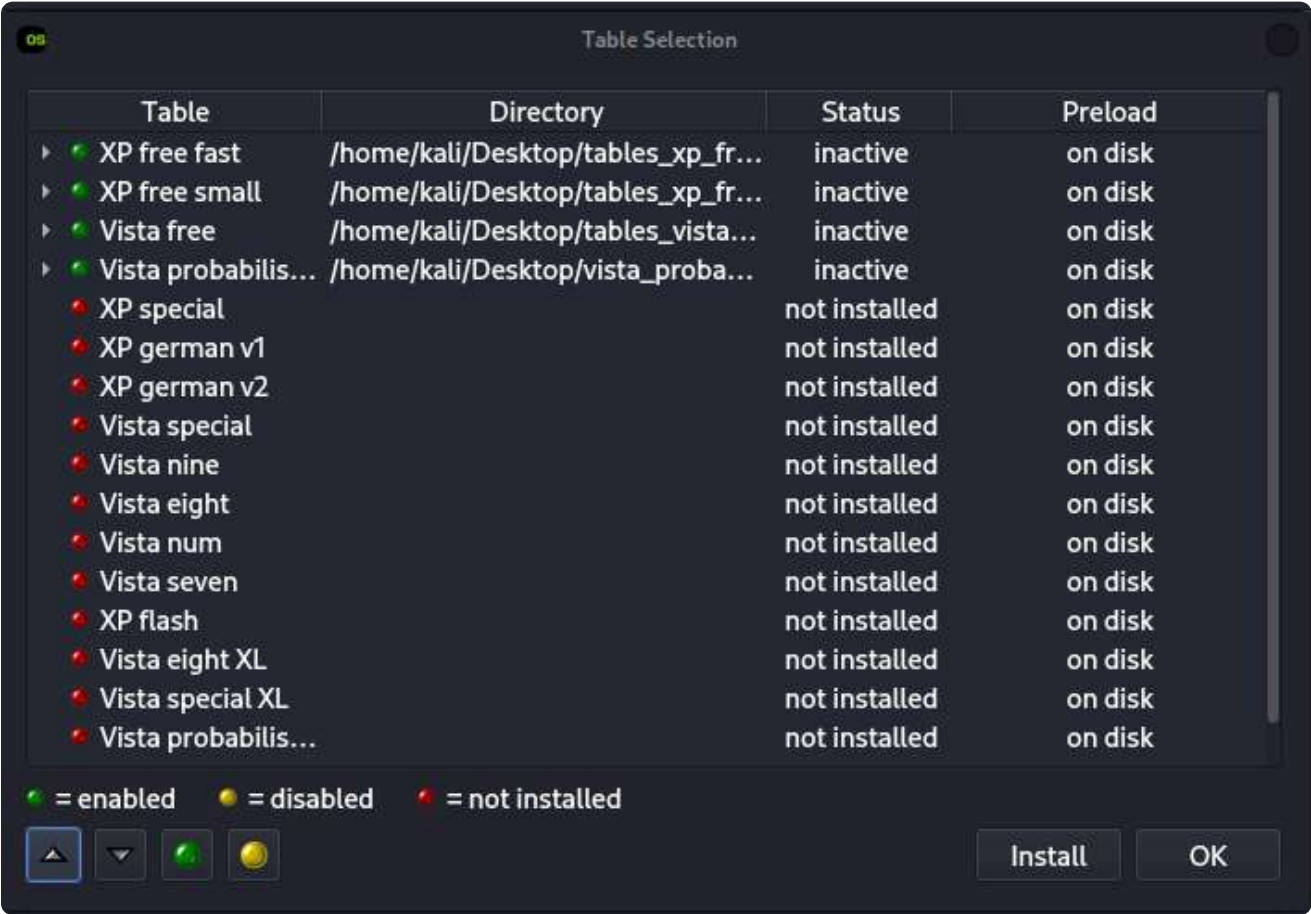
2.4

Step 1 - I download all the rainbow tables from the One-Drive that was shared to my VM with Kali. Also I downloaded the hashes that was going to be cracked (hashes_pwndump_for_you_to_crack.txt)

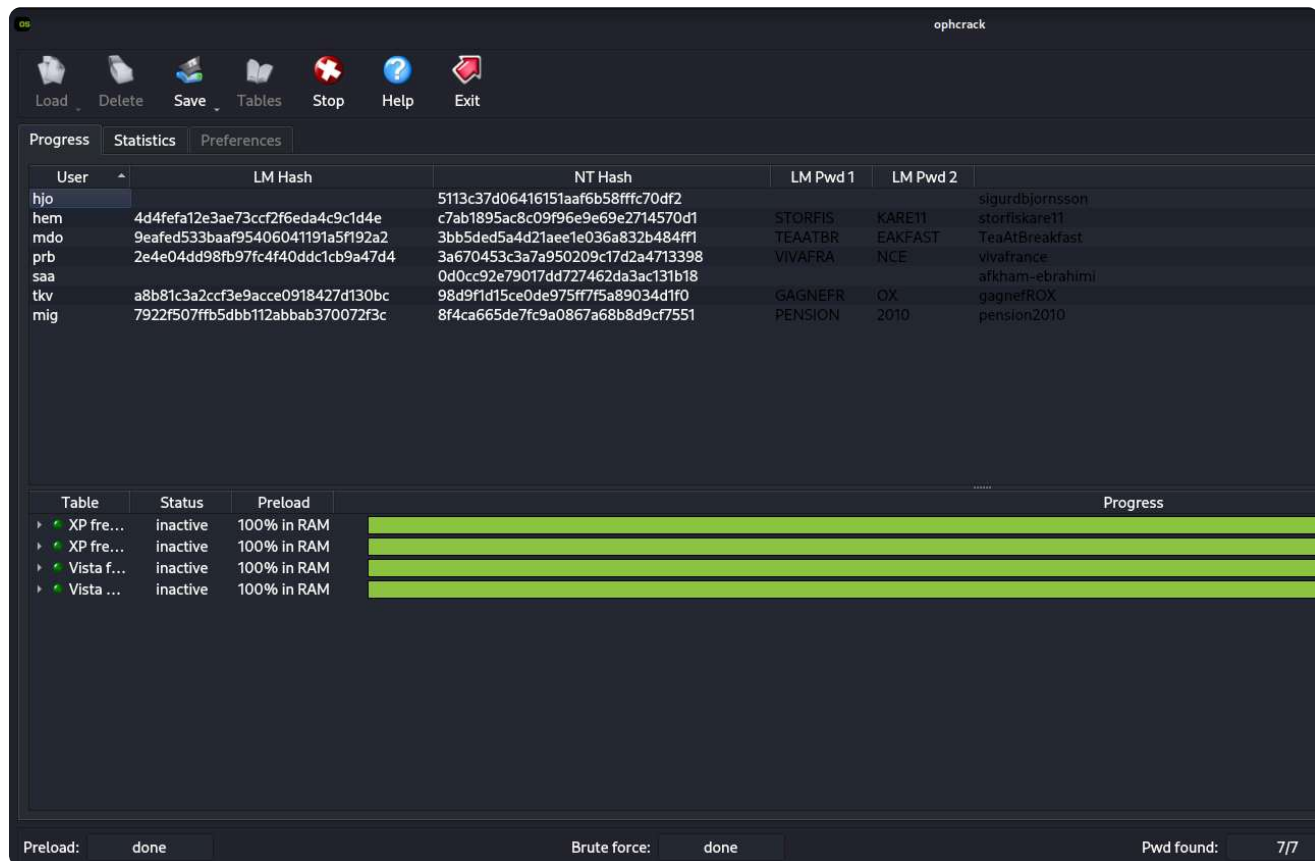
Step 2 - I launched Ophcrack and loaded the pwndump:



Step 3 - I installed the aquired rainbow tables in Ophcrack:



Step 4 - I crack function (pressed "crack") and it didnt take long until the hashes was cracked.



The LM hashes are cracked more quickly due to their weak design. LM is limited to 14 characters, splits the password into 7-character halves and is converted to uppercase. It reduces the keyspace and each half can be cracked separately.

NT keeps the case sensitivity and uses the full password. It also doesn't split, it hashes it all at once.

2.5

a) First I used "hashid -m" on the hash.txt which contains the hash "c042a646d3b38326ad0e3e8a7e12efc2".

This shows what kind of hash-type it could be:

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ hashid -m hash.txt
--File 'hash.txt'--
Analyzing 'c042a646d3b38326ad0e3e8a7e12efc2'
[+] MD2
[+] MD5 [Hashcat Mode: 0]
[+] MD4 [Hashcat Mode: 900]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000]
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600]
[+] Skype [Hashcat Mode: 23]
[+] Snefru-128
[+] NTLM [Hashcat Mode: 1000]
[+] Domain Cached Credentials [Hashcat Mode: 1100]
[+] Domain Cached Credentials 2 [Hashcat Mode: 2100]
[+] DNSSEC(NSEC3) [Hashcat Mode: 8300]
[+] RAdmin v2.x [Hashcat Mode: 9900]
--End of file 'hash.txt'--
(kali@kali)-[~/Desktop]
$
```

There after i used hashcat. I looked in the manual how to setup the command and how pick a mode and attack.

I tried first with "hashcat -m 1000 -a" together with the hash and the rockyou.txt but it was not successful.

After that i tried md5 double (mode 2600) with the command "hashcat -m 2600 -a 0"

And used "hashcat -m 2600 hash.txt -show" to see the password:

```

kali@kali: ~/Desktop

Session Actions Edit View Help

Speed.#01.....: 1273.3 kH/s (0.81ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 11046996/14344385 (77.01%)
Rejected.....: 2132/11046996 (0.02%)
Restore.Point....: 11046995/14344385 (77.01%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: JICRAK30 → JETHRO13
Hardware.Mon.#01.: Util: 66%

c042a646d3b38326ad0e3e8a7e12efc2:FandBow282926

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 2600 (md5(md5($pass)))
Hash.Target.....: c042a646d3b38326ad0e3e8a7e12efc2
Time.Started.....: Tue Jan 27 09:13:01 2026 (9 secs)
Time.Estimated...: Tue Jan 27 09:13:10 2026 (0 secs)
Kernel.Feature...: Optimized Kernel (password length 0-31 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 1270.8 kH/s (0.86ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 11173990/14344385 (77.90%)
Rejected.....: 2150/11173990 (0.02%)
Restore.Point....: 11171942/14344385 (77.88%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: Febraury. → Faith@40
Hardware.Mon.#01.: Util: 75%

Started: Tue Jan 27 09:12:43 2026
Stopped: Tue Jan 27 09:13:11 2026

(kali@kali)-[~/Desktop]
$ hashcat -m 2600 hash.txt --show
c042a646d3b38326ad0e3e8a7e12efc2:FandBow282926

(kali@kali)-[~/Desktop]
$

```

I confirmed in the rockyou.txt that the password "FandBow282926" was there also, so I would asume... I cracked it!

b) First I used the same procedure with hashid to determine what mode I needed to use. In this case I needed to try diffrent options. But with the information i had from the task I asumed it could either med NTLM or MD4. NTLM seems to be builded around MD4 i wanted to try that to.

For this task I used the context of the task on how to setup the mask for hashcat.

Also since i wanted hashcat to try several different forms with the information I had. There for I ran the full command below:

```
(kali@kali)-[~/Desktop]
$ for y in {2000..2010}; do
    hashcat -m 900 -a 3 -1 _- hash2.txt ?u?l?l?l?l?l?l?1${y} -O
done
```

First I tried with NTLM (mode 1000) without success. After that I used MD4 (mode 900) with success which shows Happier-2025 is the password:

```
kali@kali: ~/Desktop
```

```
Session  Actions  Edit  View  Help
```

```
OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
```

```
* Device #01: cpu-haswell-Intel(R) Core(TM) i7-10710U CPU @ 1.10GHz, 1469/2939 MB (512 MB allocatable), 4MCU
```

```
Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 55
```

```
INFO: All hashes found as potfile and/or empty entries! Use --show to display them.
      For more information, see https://hashcat.net/faq/potfile
```

```
Started: Tue Jan 27 13:59:28 2026
Stopped: Tue Jan 27 13:59:28 2026
hashcat (v7.1.2) starting
```

```
OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
```

```
* Device #01: cpu-haswell-Intel(R) Core(TM) i7-10710U CPU @ 1.10GHz, 1469/2939 MB (512 MB allocatable), 4MCU
```

```
Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 55
```

```
INFO: All hashes found as potfile and/or empty entries! Use --show to display them.
      For more information, see https://hashcat.net/faq/potfile
```

```
Started: Tue Jan 27 13:59:28 2026
Stopped: Tue Jan 27 13:59:29 2026
```

```
(kali@kali)-[~/Desktop]
$ hashcat -m 900 hash2.txt --show
a1a7d311c5f1417694b07cbb47d08f00:Happier-2025
```

```
(kali@kali)-[~/Desktop]
$
```

(I ran the ran hashcat before this screenshot but the PC died so i needed to reboot).

2.6

I have done some of the vulnerability scanning from previous course but I wanted to try myself.

Here I tried Nessuss and did a scan on my own host and found some

vunarbilities:

tenable

Nessus Expert

ScansSettings

bitdestroyer

FOLDERS

My Scans

All Scans

Trash

RESOURCES

Policies

Plugin Rules

Customized Reports

Myhost

ConfigureAudit TrailLaunchReportExport

Scan SummaryHosts 1Vulnerabilities 41Remediations 8History 1

FilterSearch Hosts1 Host

HostAuthVulnerabilities

127.0.0.1Pass

222351

Scan Details

Policy:Basic Network Scan

Status:Completed

Severity Base:CVSS v3.0

Scanner:Local Scanner

Start:Today at 11:35 AM

End:Today at 11:41 AM

Elapsed:6 minutes

Vulnerabilities

Critical

High

Medium

tenable

Nessus Expert

ScansSettings

bitdestroyer

FOLDERS

My Scans

All Scans

Trash

RESOURCES

Policies

Plugin Rules

Customized Reports

Vulnerabilities 41

FilterSearch Vulnerabilities41 Vulnerabilities

SevCVSSVPR EPSSFamilyCount

MIXED... ..Ubuntu Local Security Checks8

MIXED... ..General4

INFO... ..General6

INFO... ..Web Servers2

INFO... ..General2

INFO... ..Service detection2

INFO... ..Service detection2

INFO... ..General1

Host Details

127.0.0.1

E6:59:0E:74:DF:22

Linux Kernel 6.6.87.2-microsoft-standard-WSL2 on Ubuntu 24.04

Today at 5:35 AM

Today at 5:41 AM

6 minutes

Download

Pass

Vulnerabilities

Critical

High

Medium

Low

Info

tenable

Nessus Expert

ScansSettings

bitdestroyer

FOLDERS

My Scans

All Scans

Trash

RESOURCES

Policies

Plugin Rules

Customized Reports

Search Vulnerabilities8 Vulnerabilities

SevCVSSVPR EPSSFamilyCount

CRITICAL9.16.00.0028Ubuntu Local Security Checks1

CRITICAL9.16.00.0004Ubuntu Local Security Checks1

HIGH7.16.70.0004Ubuntu Local Security Checks1

HIGH7.07.40.0001Ubuntu Local Security Checks1

MEDIUM6.54.40.0003Ubuntu Local Security Checks1

LOW3.76.00.002Ubuntu Local Security Checks1

LOW3.72.20.0004Ubuntu Local Security Checks1

LOW3.34.40.0014Ubuntu Local Security Checks1

Host Details

Policy:Basic Network Scan

Status:Completed

Severity Base:CVSS v3.0

Scanner:Local Scanner

Start:Today at 11:35 AM

End:Today at 11:41 AM

Elapsed:6 minutes

Vulnerabilities

Critical

High

Medium

Low

Info

When using docker scout:

https://hackmd.io/6HBneBQ3SruMyNJSHGycrg?view

18/21

```
Administrator: Kommandotolken
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. Med ensamrätt.

C:\Windows\System32>docker scout version

          _____
         /  _  /  _  /
        /  /  /  /  /
       /  /  /  /  /
      /  /  /  /  /
     /  /  /  /  /
    /  /  /  /  /
   /  /  /  /  /
  /  /  /  /  /
 /  /  /  /  /
/  /  /  /  /

 Docker
Scout

version: v1.19.0 (go1.25.0 - windows/amd64)
git commit: 91fd9f3262322a34537a7a7bbcd5cfd282b93402
```

```
Administrator: Kommandotolken

C:\Windows\System32>docker pull nginx:1.18
1.18: Pulling from library/nginx
f7ec5a41d630: Pull complete
1576642c9776: Pull complete
03f221d9cf00: Pull complete
0b20d28b5eb3: Pull complete
c12a848bad84: Pull complete
Digest: sha256:e90ac5331fe095cea01b121a3627174b2e33e06e83720e9a934c7b8ccc9c55a0
Status: Downloaded newer image for nginx:1.18
docker.io/library/nginx:1.18

C:\Windows\System32>docker scout cves
  ✓ Image stored for indexing
  ✓ Indexed 241 packages
  ! failed to delete temporary image archive C:\Users\A\AppData\Local\Temp\docker-scout\sha256\0bb571b2299ff36bbae4ce85b439cb05a4a29cbaab241948897c20aa0e688b71\7fca548b-11b3-4fca-b0f2-1a7835bf8efa: unlinkat C:\Users\A\AppData\Local\Temp\docker-scout\sha256\0bb571b2299ff36bbae4ce85b439cb05a4a29cbaab241948897c20aa0e688b71\7fca548b-11b3-4fca-b0f2-1a7835bf8efa: The process cannot access the file because it is being used by another process.  ✗ Detected 19 vulnerable packages with a total of 44 vulnerabilities

## Overview

| Analyzed Image
```



```
Administrator: Kommandotolken

Target
digest      0bb571b2299f
platform    linux/amd64
vulnerabilities 0C 2H 23M 19L
size        394 MB
packages     241

## Packages and Vulnerabilities

0C 1H 1M 1L gnupg2 2.4.4-2ubuntu17.3
pkg:deb/ubuntu/gnupg2@2.4.4-2ubuntu17.3?os_distro=noble&os_name=ubuntu&os_version=24.04

x HIGH CVE-2025-68973
https://scout.docker.com/v/CVE-2025-68973
Affected range : <2.4.4-2ubuntu17.4
Fixed version  : 2.4.4-2ubuntu17.4
CVSS Score     : 7.0
CVSS Vector    : CVSS:3.1/AV:L/AC:H/PR:L/UI:N/S:U/C:H/I:H/A:H

x MEDIUM CVE-2025-68972
https://scout.docker.com/v/CVE-2025-68972
Affected range : >=0
Fixed version  : not fixed
CVSS Score     : 4.7
CVSS Vector    : CVSS:3.1/AV:L/AC:H/PR:L/UI:N/S:U/C:N/I:H/A:N

x LOW CVE-2022-3219
```